JSPM University Pune
Faculty of Science and Technology
School of Computational Sciences
M.Sc. Sem-2
Data Science and Artificial Intelligence (DSAI)/ Cyber Security (C.S.)

EXPERIMENT NO: 1 **DATE: / /**

AIM : INTRODUCTION TO VIRTUAL PLATFORM WOKWI (HTTPS://WOKWI.COM/)

OBJECTIVE: The main objective of this experiment is,

A. Create an account on Wokwi
B. Identify the different devices and measuring instrument.
C. Simulate a simple circuit in Wokwi

DESCRIPTION:

Wokwi is a browser-based online electronics and IoT simulator that allows users to build, code, and test embedded systems projects in a virtual environment. It supports popular boards like Arduino, ESP32, STM32, and Raspberry Pi Pico, along with a wide range of virtual components.

Key features:

- Browser-based: No installations or downloads are required, allowing users to start simulating directly from any modern web browser.
- Virtual hardware: Users can add and wire components virtually, such as LEDs, sensors, displays, and motors, and interact with them in real-time.
- Code editor: A built-in code editor lets you write and edit code directly in the platform, with support for languages like C++ (for Arduino) and MicroPython.
- IoT capabilities: The simulator includes virtual Wi-Fi and networking features, allowing you to test internet-connected projects using protocols like MQTT and HTTP.
- VS Code integration: A Visual Studio Code extension is available, enabling embedded and IoT developers to integrate Wokwi's simulation directly into their local development workflow.
- Debugging tools: Advanced tools like a virtual logic analyzer and GDB integration help in debugging code and analyzing digital signals.

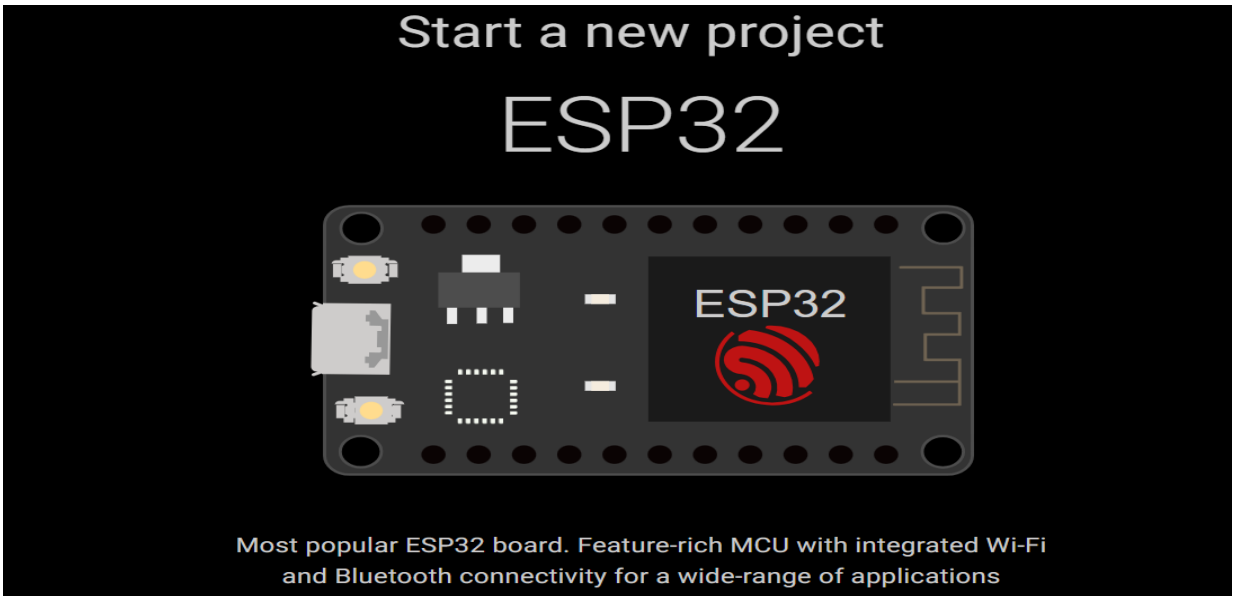
- Collaboration: Projects can be easily shared with others via a simple link, which is useful for remote learning and team projects.

Common uses:

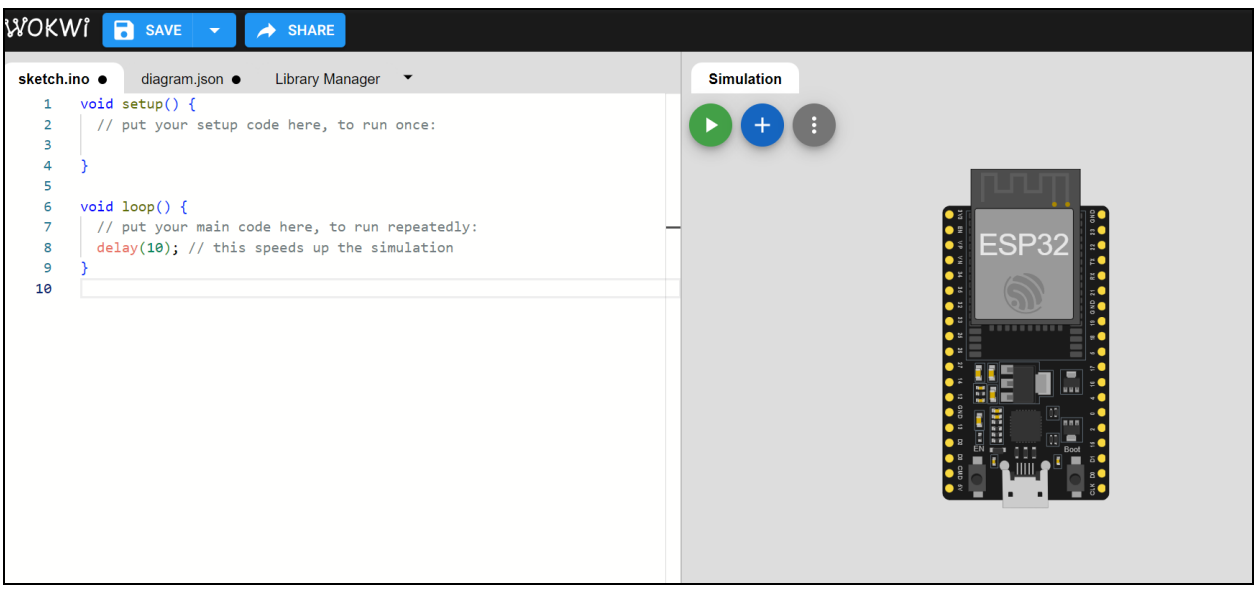
- Education: It serves as a teaching tool for students to learn about microcontrollers, electronics, and programming without needing physical hardware.
- Prototyping: Developers and hobbyists can quickly test and iterate on project ideas, saving time and money on components.
- Debugging: Users can diagnose issues with code and circuit design in a virtual environment before moving to physical hardware.
- Development: It is used for developing and testing firmware for embedded and IoT applications.

STEPS TO CREATE NEW PROJECT:

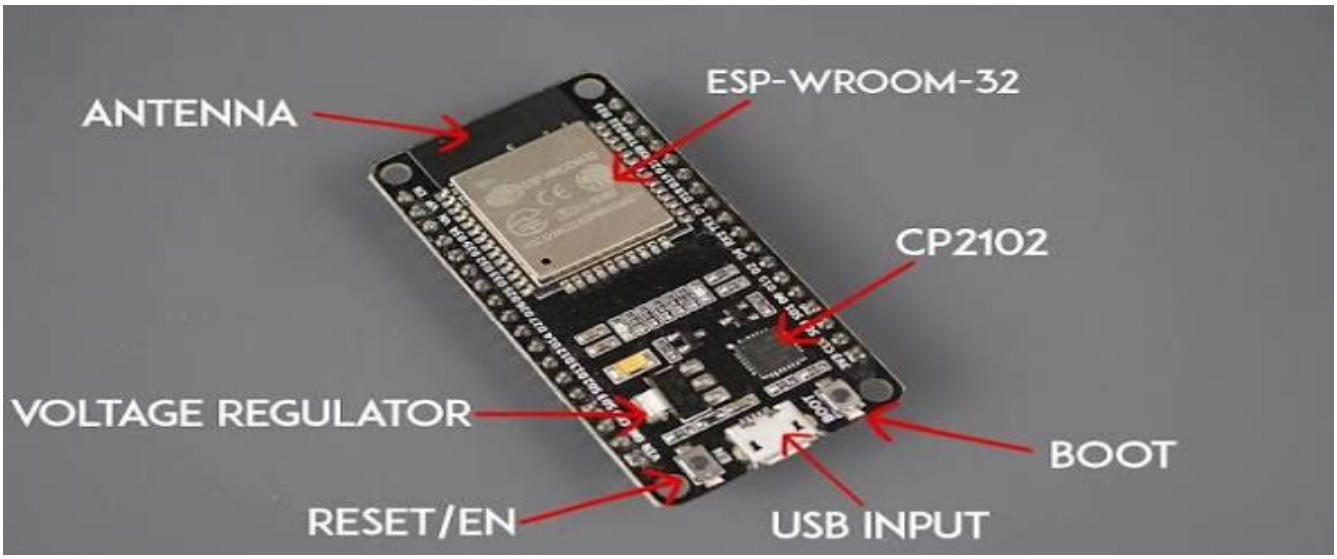
To get started, navigate to the Wokwi Simulator website. From there, you'll see the following dashboard. You can use the app for free and without creating a user account.



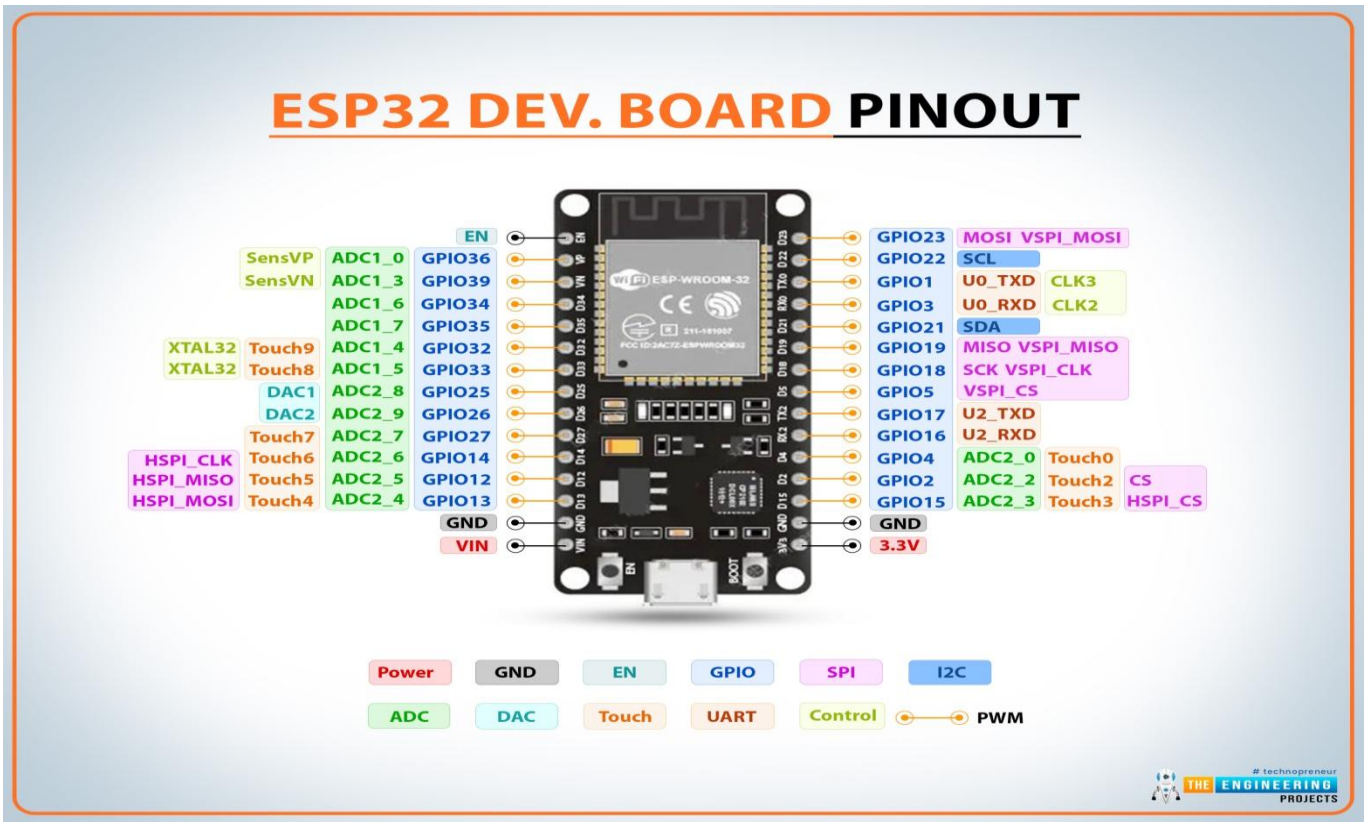
Wokwi Main Sheet



The ESP32 is a powerful and versatile system-on-a-chip (SoC) microcontroller, known for its integrated Wi-Fi and Bluetooth capabilities. This makes it a popular choice for Internet of Things (IoT) applications, embedded systems, and other projects requiring wireless connectivity



ESP32 PIN Diagram :-



Key features:

<u>Processing and memory</u>

- Dual-core or single-core processor: The standard ESP32 features a dual-core, 32-bit Tensilica Xtensa LX6 microprocessor, with some variants like the ESP32-S0WD using a

single-core version. Newer ESP32-C series models use a single-core RISC-V CPU, and ESP32-S3 uses a dual-core Xtensa LX7 CPU with machine learning acceleration.

- Adjustable clock frequency: The clock speed is adjustable from 80 MHz to 240 MHz, allowing developers to balance performance and power consumption.
- Ultra-low-power (ULP) co-processor: For extended battery life, the ULP co-processor can perform simple tasks like sensor monitoring while the main CPU is in deep sleep.
- Internal memory: Memory includes 520 KB of on-chip SRAM for data and instructions, plus additional ROM and RTC SRAM for boot and low-power operations.
- Expandable flash memory: Modules typically come with 4 MB of external flash memory for storing programs and data, with some variants supporting additional external PSRAM.

Wireless connectivity

- Wi-Fi: Supports 802.11 b/g/n Wi-Fi, allowing the ESP32 to connect to routers in Station mode, create its own network in Access Point (SoftAP) mode, or operate in both simultaneously.
- Bluetooth: Features dual-mode Bluetooth connectivity with support for both Classic Bluetooth (BR/EDR) and Bluetooth Low Energy (BLE), enabling it to communicate with a wide range of devices.
- Wi-Fi 6 (802.11ax): Newer variants, such as the ESP32-C6, also feature Wi-Fi 6 support.

Power management

- Multiple low-power modes: The ESP32 is designed for energy efficiency, offering several power-saving modes to optimize battery life.
- Modem-sleep: Wi-Fi and Bluetooth are disabled, but the CPU remains active. Power consumption is ~3–20 mA.
- Light-sleep: The CPU and most RAM are clock-gated, but the Wi-Fi/Bluetooth connection is maintained. Current consumption is approximately 0.8 mA.
- Deep-sleep: Powers down the CPU and most RAM, with only the real-time clock (RTC) and ULP co-processor remaining active. This dramatically reduces power consumption, making it ideal for battery-operated devices.
- Hibernation: The most extreme power-saving mode, where nearly all hardware is turned off, with consumption as low as 5 µA.

Peripherals and interfaces

- Extensive GPIOs: Offers numerous programmable General-Purpose Input/Output (GPIO) pins for connecting to a wide variety of sensors and peripherals.
- Analog-to-Digital Converter (ADC): Features multiple 12-bit SAR ADC channels for reading analog sensor data.
- Digital-to-Analog Converter (DAC): Includes 8-bit DAC channels for converting digital signals to analog voltages.
- Capacitive touch sensors: Contains built-in touch sensors, which can be used to create touch-sensitive pads without external hardware.
- Common communication protocols: Supports standard interfaces such as SPI, I^2C, I^2S, UART, and PWM, allowing for flexible connectivity.
- Other peripherals: Depending on the model, additional peripherals can include a Hall sensor, Ethernet MAC interface, and an SD/SDIO/MMC Host controller.

Security

- Secure boot: Ensures that only trusted and authenticated software can execute on the device, preventing untrustworthy code from running.
- Flash encryption: Encrypts the contents of the external flash memory, protecting intellectual property and sensitive data.
- Hardware cryptographic acceleration: Features hardware acceleration for cryptographic algorithms like AES, SHA-2, and RSA, ensuring secure communication.
- Secure OTA updates: Allows for secure Over-the-Air firmware updates, ensuring that new code is authenticated and encrypted during transit.

- RESULTS:

- CONCLUSION:

Sample Viva Questions

1) What is the Wokwi simulator, and how does it work for simulating microcontroller projects?

2) What are the key differences between simulating a project on Wokwi versus testing on actual hardware (e.g., ESP32 or Arduino)?